Ex. No : 11 Date : 03/09/2024

Routing a Network layer..

a) Simulate static Routing protocal configuration
b) Simulate RIP

Aim:

To simulate static Routing protocol configuration and RIP using Cisco Packet Tracer.

Algorithm / Procedure:

PART - A (Static Routing):

*. Open Cisco Packet Tracer and create the given network topology with routers.

*. configure the networks available on the router interfaces.

*. Open the router CLI and enter the required configuration mode.

*. configure the first route as the main route and the second route as the backup route.

*. Use the tracert command to trace the path taken by the Packets.

*. To test the backup route, remove the link between the routers and observe the change in the routing path.

*. configure the backup static route and verify that the packet is transmitted through the alternate route.

PART B - RIP using cisco packet tracer.

*. configure the IP addresses of the PCs and router interfaces according to the given network.

*. configure the serial interfaces of the routers with the given IP addresses ~~configuration mode~~.

*. configure and enable the RIP routing Protocol from global configuration mode.

*. specify the directly connected networks that have to be advertised using RIP.

*. use the ping command to test connectivity between the PCs.

*. use the tracert command to observe the route followed by the packets.

*. If the selected route fails, RIP automatically selects another available route for packet transmission.

11.a

```
Router>enable
Router#show ip route static
S     30.0.0.0/8 [20/0] via 40.0.0.2
S     50.0.0.0/8 [10/0] via 40.0.0.2
S     60.0.0.0/8 [1/0] via 40.0.0.2

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#show ip route static
S     30.0.0.0/8 [10/0] via 20.0.0.2
S     50.0.0.0/8 [10/0] via 40.0.0.2
S     60.0.0.0/8 [1/0] via 40.0.0.2
```

```
Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=55ms TTL=125
Reply from 20.0.0.2: bytes=32 time=56ms TTL=125
Reply from 20.0.0.2: bytes=32 time=60ms TTL=125
Reply from 20.0.0.2: bytes=32 time=55ms TTL=125

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 55ms, Maximum = 60ms, Average = 56ms

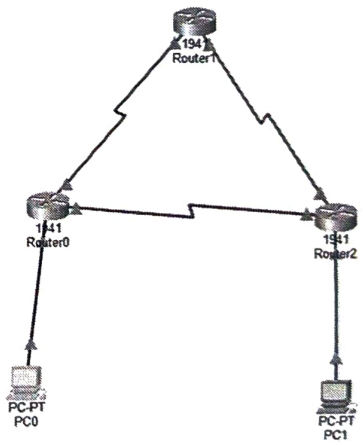
C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2   11 ms      8 ms      5 ms      192.168.1.250
  3   29 ms     26 ms     25 ms      192.168.1.245
  4   22 ms     28 ms     20 ms      20.0.0.2

Trace complete.
```

11.b



11.b

Result:

Thus static routing & RIP were successfully simulated using Cisco Packet tracer and packet transmission and alternate routing were verified.